

Symmetric tensor categories

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Lecture 1

The model example: $\text{Rep}_{\mathbb{k}}(G)$

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We would now like to axiomatize the categorical structures visible in $\text{Rep}_{\mathbb{k}}(G)$ without referring to underlying vector spaces.

Monoidal category: data

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with the following properties.

Monoidal category: pentagon axiom

First, the associativity constraint is required to satisfy the **pentagon axiom**: for all $W, X, Y, Z \in \mathcal{C}$, the diagram

$$\begin{array}{ccc} ((W \otimes X) \otimes Y) \otimes Z & \xrightarrow{a_{W \otimes X, Y, Z}} & (W \otimes X) \otimes (Y \otimes Z) \\ a_{W, X, Y} \otimes \text{id}_Z \downarrow & & \downarrow a_{W, X, Y \otimes Z} \\ (W \otimes (X \otimes Y)) \otimes Z & & W \otimes (X \otimes (Y \otimes Z)) \\ a_{W, X \otimes Y, Z} \downarrow & \nearrow \text{id}_W \otimes a_{X, Y, Z} & \\ W \otimes ((X \otimes Y) \otimes Z) & & \end{array}$$

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commutes. Equivalently,

$$a_{W, X, Y \otimes Z} a_{W \otimes X, Y, Z} = (\text{id}_W \otimes a_{X, Y, Z}) a_{W, X \otimes Y, Z} (a_{W, X, Y} \otimes \text{id}_Z).$$

Monoidal category: unit axiom

Next, the unit object $\mathbf{1}$ is required to satisfy

$$\mathbf{1} \otimes \mathbf{1} \cong \mathbf{1},$$

and tensor multiplication by $\mathbf{1}$ on either side must be an equivalence:

$$L_{\mathbf{1}} : \mathcal{C} \rightarrow \mathcal{C}, \quad X \mapsto \mathbf{1} \otimes X,$$

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The **MacLane's coherence theorem** implies that we may ignore parentheses and unit objects inside tensor products.

Braided monoidal category: data

A **braided monoidal category** is a monoidal category $\mathcal{C}$ together with a natural isomorphism

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They imply that for every object $X \in \mathcal{C}$ the object $X^{\otimes n}$ carries a natural action of the **braid group** B_n , with generator b_i acting by c in the i -th and $i+1$ -th factors.

Braided monoidal category: hexagon axioms

1. For all $X, Y, Z \in \mathcal{C}$, the diagram

$$\begin{array}{ccccc} (X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) & \xrightarrow{c_{X,Y \otimes Z}} & (Y \otimes Z) \otimes X \\ \downarrow c_{X,Y} \otimes \text{id}_Z & & & & \downarrow a_{Y,Z,X} \\ (Y \otimes X) \otimes Z & \xrightarrow{a_{Y,X,Z}} & Y \otimes (X \otimes Z) & \xrightarrow{\text{id}_Y \otimes c_{X,Z}} & Y \otimes (Z \otimes X) \end{array}$$

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2. For all $X, Y, Z \in \mathcal{C}$, the diagram

$$\begin{array}{ccccc} X \otimes (Y \otimes Z) & \xrightarrow{a_{X,Y,Z}^{-1}} & (X \otimes Y) \otimes Z & \xrightarrow{c_{X \otimes Y,Z}} & Z \otimes (X \otimes Y) \\ \text{id}_X \otimes c_{Y,Z} \downarrow & & & & \uparrow a_{Z,X,Y} \\ X \otimes (Z \otimes Y) & \xrightarrow{a_{X,Z,Y}^{-1}} & (X \otimes Z) \otimes Y & \xrightarrow{c_{X,Z} \otimes \text{id}_Y} & (Z \otimes X) \otimes Y \end{array}$$

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Symmetric monoidal category

A braided monoidal category is **symmetric** if the braiding is involutive:

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In $\text{Rep}_{\mathbb{k}}(G)$, c_{VW} is the flip

$$v \otimes w \mapsto w \otimes v.$$

Rigidity: dual data

Let $\mathcal{C}$ be monoidal. A **left dual** of $X \in \mathcal{C}$ is an object X^* with morphisms

$$\mathrm{ev}_X : X^* \otimes X \rightarrow \mathbf{1}, \quad \mathrm{coev}_X : \mathbf{1} \rightarrow X \otimes X^*,$$

satisfying the two triangular identities.

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A **right dual** is defined analogously, with evaluation $X \otimes {}^*X \rightarrow \mathbf{1}$ and coevaluation $\mathbf{1} \rightarrow {}^*X \otimes X$. A monoidal category is **rigid** if every object has both a left and a right dual.

Rigidity: triangular identities

The left-dual triangular identities are the commutativity of the following two diagrams:

$$\begin{array}{c} X \xrightarrow{\text{coev}_X \otimes \text{id}_X} (X \otimes X^*) \otimes X \xrightarrow{a_{X, X^*, X}} X \otimes (X^* \otimes X) \xrightarrow{\text{id}_X \otimes \text{ev}_X} X \\ \searrow \text{id}_X \swarrow \\ X^* \xrightarrow{\text{id}_{X^*} \otimes \text{coev}_X} X^* \otimes (X \otimes X^*) \xrightarrow{a_{X^*, X, X^*}^{-1}} (X^* \otimes X) \otimes X^* \xrightarrow{\text{ev}_X \otimes \text{id}_{X^*}} X^* \end{array}$$

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Note that if X^* or *X exists for a given X then it is unique up to a unique isomorphism, and that in a symmetric category we have a canonical isomorphism ${}^*X \cong X^*$, so there is only one dual.

Rigidity in $\text{Rep}_{\mathbb{k}}(G)$

For $V \in \text{Rep}_{\mathbb{k}}(G)$, set

$$V^* = \text{Hom}_{\mathbb{k}}(V, \mathbb{k}),$$

with contragredient G -action. Evaluation and coevaluation are

$$\text{ev} : V^* \otimes V \rightarrow \mathbb{k}, \quad \varphi \otimes v \mapsto \varphi(v),$$

$$\text{coev} : \mathbb{k} \rightarrow V \otimes V^*, \quad 1 \mapsto \sum_i v_i \otimes v_i^*,$$

for a basis $\{v_i\}$ and its dual basis $\{v_i^*\}$.

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for a basis $\{v_i\}$ and its dual basis $\{v_i^*\}$. These maps are G -equivariant and satisfy the triangular identities.

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- $\text{End}(\mathbf{1}) = \mathbb{k}$;

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A braided tensor category is called **symmetric** if its braiding is symmetric.

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It is a whole world where we can do algebra and representation theory!

Examples of symmetric tensor categories

For example, $\text{Rep}_{\mathbb{k}}(G)$ is a symmetric tensor category. The most basic case is $G = 1$, which gives the category of finite dimensional $\mathbb{k}$ -vector spaces, $\text{Vec}_{\mathbb{k}}$.

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An affine group scheme over $\mathbb{k}$ is a group object G in the category of affine schemes over $\mathbb{k}$. Concretely, it is an affine $\mathbb{k}$ -scheme G whose algebra of regular functions $\mathcal{O}(G)$ is endowed with a structure of a commutative Hopf algebra.

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A **rational representation** of an affine group scheme G is a finite-dimensional $\mathcal{O}(G)$ -comodule V , i.e., a vector space V with a **coaction**

$$\rho : V \rightarrow V \otimes \mathcal{O}(G)$$

which defines an action of the algebra $\mathcal{O}(G)^*$ on V . It is easy to see that rational representations of an affine group scheme G form a symmetric tensor category $\text{Rep}(G)$.

Tensor functors

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Let $\mathcal{C}, \mathcal{D}$ be monoidal categories. A **monoidal functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of a functor and coherent isomorphisms

$$J_{X,Y} : F(X) \otimes F(Y) \xrightarrow{\sim} F(X \otimes Y), \quad \nu : \mathbf{1}_{\mathcal{D}} \xrightarrow{\sim} F(\mathbf{1}_{\mathcal{C}}),$$

compatible with the associativity and unit data. A **braided (symmetric) monoidal functor** is a monoidal functor between braided (symmetric) monoidal categories compatible with the braidings.

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If $\mathcal{C}, \mathcal{D}$ are tensor categories then a monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called a **tensor functor** if it is $\mathbb{k}$ -linear and faithful (hence exact). A **fiber functor** is a symmetric tensor functor to the target $\mathbf{Vec}_{\mathbb{k}}$.

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Thus we would like to explain why the previous examples **are equivalent** to $\mathbf{Rep}(G)$ for some affine group scheme G as symmetric tensor categories.

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This is a **fiber functor**: a symmetric tensor functor to $\text{Vec}_{\mathbb{k}}$.

Tensor automorphisms of a fiber functor

Let $\omega : \mathcal{C} \rightarrow \text{Vec}_{\mathbb{k}}$ be a fiber functor. Define the affine group functor

$$\underline{\text{Aut}}_{\otimes}(\omega) : R \mapsto \text{Aut}^{\otimes}(\omega_R), \quad \omega_R(X) = \omega(X) \otimes_{\mathbb{k}} R.$$

Thus an R -point of $\underline{\text{Aut}}_{\otimes}(\omega)$ is a family

$$\{g_X : \omega(X) \otimes R \xrightarrow{\sim} \omega(X) \otimes R\}_{X \in \mathcal{C}}$$

which is natural in X , preserves tensor products and the unit, and is invertible.

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For $\mathcal{C} = \text{Rep}(G)$ and $\omega = \omega_G$, evaluation of representations gives

$$G \cong \underline{\text{Aut}}_{\otimes}(\omega_G).$$

The Coend of matrix coefficients

The functor $\underline{\text{Aut}}_{\otimes}(\omega)$ is represented by a Hopf algebra constructed from matrix coefficients. Set

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Concretely, this is the quotient of

$$\bigoplus_{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X)$$

by the relations

$$\omega(f)^* \varphi \otimes v - \varphi \otimes \omega(f)v \in \omega(Y)^* \otimes \omega(Y)$$

for all morphisms $f : X \rightarrow Y$, where $\varphi \in \omega(Y)^*$ and $v \in \omega(X)$.

Hopf algebra structure on the Coend

Write the class of $\psi \otimes v \in \omega(X)^* \otimes \omega(X)$ as $[\psi \mid v]_X$. The coalgebra structure is the usual one for matrix coefficients:

$$\Delta([\psi \mid v]_X) = \sum_i [\psi \mid e_i]_X \otimes [e_i^* \mid v]_X, \quad \varepsilon([\psi \mid v]_X) = \psi(v),$$

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The tensor structure of ω defines multiplication:

$$[\psi \mid v]_X [\varphi \mid w]_Y = [\psi \otimes \varphi \mid v \otimes w]_{X \otimes Y},$$

after identifying $\omega(X \otimes Y) \cong \omega(X) \otimes \omega(Y)$. Duals give the antipode.

Tannakian reconstruction

Let $\mathcal{C}$ be a symmetric tensor category and let

$$\omega : \mathcal{C} \rightarrow \mathbf{Vec}_{\mathbb{k}}$$

be a fiber functor. Put

$$G := \underline{\mathbf{Aut}}_{\otimes}(\omega), \quad \mathcal{O}(G) = \int^{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X).$$

Then the canonical coactions on $\omega(X)$ define a symmetric tensor equivalence

$$\mathcal{C} \simeq \mathbf{Rep}(G).$$

This shows that all our examples are of the form $\mathbf{Rep}(G)$ for an affine group scheme G .

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It was shown by Deligne and Milne that the fiber functor is unique up to isomorphism; consequently G is uniquely determined up to conjugation.

Abstract groups and proalgebraic completion

Let us now explain how this works in our first example. Let Γ be an abstract group. The forgetful functor gives

$$\widehat{\Gamma} := \underline{\mathrm{Aut}}_{\otimes}(\omega_{\Gamma}), \quad \mathrm{Rep}_{\mathbb{k}}(\Gamma) \simeq \mathrm{Rep}(\widehat{\Gamma}).$$

The affine group scheme $\widehat{\Gamma}$ is the **proalgebraic completion** of Γ over $\mathbb{k}$.

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Its coordinate algebra is the subalgebra of $\mathrm{Fun}(\Gamma, \mathbb{k})$ spanned by the matrix coefficients

$$g \mapsto \langle f, gv \rangle, \quad v \in V, \quad f \in V^*, \quad V \in \mathrm{Rep}_{\mathbb{k}}(\Gamma).$$

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Thus the group scheme $\widehat{\Gamma}$ is reduced, i.e., it is a proalgebraic group. Namely, it is the inverse limit of algebraic groups G equipped with a homomorphism from Γ , i.e.,

$$\widehat{\Gamma} = \varprojlim_{\Gamma \rightarrow G} G.$$

In particular, $\widehat{\Gamma}$ carries a canonical (universal) homomorphism $\Gamma \rightarrow \widehat{\Gamma}$ (which may or may not be injective).

Tannakian categories

A symmetric tensor category $\mathcal{C}$ over $\mathbb{k}$ is called **Tannakian** if it admits a fiber functor

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Equivalently, by reconstruction,

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Thus Tannakian categories are precisely the symmetric tensor categories whose objects can be realized as finite-dimensional vector spaces, functorially and compatibly with tensor products, duals, and the symmetry.

So it is natural to ask: **are there other examples of symmetric tensor categories?**

The first non-Tannakian example - supervector spaces

The answer turns out to be **yes**. Assume $\text{char } \mathbb{k} \neq 2$. Let $\text{sVec}_{\mathbb{k}}$ be the category of finite-dimensional **supervector spaces**

$$V = V_{\bar{0}} \oplus V_{\bar{1}}.$$

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The underlying tensor category is equivalent to $\text{Rep}_{\mathbb{k}}(\mathbb{Z}/2)$, but the symmetric structure is different. But why is it a genuinely new example?

Trace and categorical dimension

Let X be an object of a symmetric tensor category and let $f : X \rightarrow X$. The **categorical trace** is

$$\mathrm{Tr}(f) : \mathbf{1} \xrightarrow{\mathrm{coev}_X} X \otimes X^* \xrightarrow{f \otimes 1} X \otimes X^* \xrightarrow{c_{X, X^*}} X^* \otimes X \xrightarrow{\mathrm{ev}_X} \mathbf{1}.$$

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In $\mathrm{Vec}_{\mathbb{k}}$, the trace is the usual trace and dimension is the usual dimension viewed as an element of $\mathbb{k}$. In $\mathrm{sVec}_{\mathbb{k}}$, the trace is the **supertrace** $\mathrm{Tr}(f) = \mathrm{Tr}(f_0) - \mathrm{Tr}(f_1)$, and the dimension is the **superdimension**

$$\dim(V) = \dim_{\mathbb{k}} V_{\bar{0}} - \dim_{\mathbb{k}} V_{\bar{1}}.$$

Invertible objects distinguish $\mathbf{sVec}$ from Tannakian categories

An object X in a monoidal category is **invertible** if there exists Y such that

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In $s\mathbf{Vec}_{\mathbb{k}}$, the one-dimensional odd line $\mathbb{k}^{0|1}$ is invertible and satisfies

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How can we generalize this example?

Affine supergroup schemes

Affine supergroup schemes

An **affine superscheme** X over $\mathbb{k}$ is represented by a commutative algebra $\mathcal{O}(X) = \mathcal{O}(X)_{\bar{0}} \oplus \mathcal{O}(X)_{\bar{1}}$ in $\mathbf{sVec}_{\mathbb{k}}$ (a.k.a. **supercommutative algebra**).

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Equivalently, G is a representable group-valued functor of points on supercommutative $\mathbb{k}$ -algebras.

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Equivalently, G is a representable group-valued functor of points on supercommutative $\mathbb{k}$ -algebras.

Basic examples of affine supergroup schemes are classical algebraic supergroups, such as $GL(m|n)$ and $Osp(m|2n)$.

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Equivalently, G is a representable group-valued functor of points on supercommutative $\mathbb{k}$ -algebras.

Basic examples of affine supergroup schemes are classical algebraic supergroups, such as $GL(m|n)$ and $Osp(m|2n)$.

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There is a slight generalization of this example. Suppose $z \in G(\mathbb{k})$ is an element of order 2 acting on $\mathcal{O}(G)$ by parity. Then we may consider the category $\mathcal{C} \simeq \mathbf{Rep}(G, z)$ of representations of G on which z acts by parity. This is also a symmetric tensor category, and $\mathbf{Rep}(G) \cong \mathbf{Rep}(\{1, z\} \ltimes G, z)$, so it generalizes the previous example.

Superfiber functors and super-Tannakian categories

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Super-Tannakian reconstruction

Let

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be a superfiber functor. Deligne showed that in this case $\mathcal{C} = \mathbf{Rep}(G, z)$ for a unique pair (G, z) up to conjugation.

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The coordinate Hopf superalgebra is again a Coend of matrix coefficients:

$$\mathcal{O}(G) = \int^{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X),$$

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Since ω is unique, so is (G, z) .

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We have thus enlarged the Tannakian class from fiber functors to $\mathbf{Vec}_{\mathbb{k}}$ to fiber functors to $\mathbf{sVec}_{\mathbb{k}}$, i.e., to super-Tannakian categories. The next question is:

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However, the answer in characteristic zero becomes **yes** under the condition of **moderate growth**.

Moderate growth

Let $\mathcal{C}$ be a tensor category. For $X \in \mathcal{C}$, let

$$\ell_n(X) := \ell(X^{\otimes n})$$

be the length of the n -th tensor power. The category $\mathcal{C}$ has **moderate growth** if for every X there exists $C_X > 0$ such that

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Thus every super-Tannakian category has moderate growth.

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Let $\mathbb{k}$ be algebraically closed of characteristic 0. A symmetric tensor category over $\mathbb{k}$ has moderate growth if and only if it is super-Tannakian.

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Equivalently, under moderate growth there exists a superfiber functor

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This shows that while we can do algebra in any symmetric tensor category, in characteristic zero and moderate growth setting such algebra can be described in classical terms, as ordinary equivariant algebra under a supergroup scheme.

Lecture 2

Deligne categories without moderate growth

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The example is Deligne's interpolation $\text{Rep } GL_t$ of $\text{Rep}_{\mathbb{C}} GL_n$ in which n is replaced by a parameter $t \in \mathbb{C}$.

For $t \notin \mathbb{Z}$, the category $\text{Rep } GL_t$ is semisimple symmetric tensor category, but it is not super-Tannakian because it does not have moderate growth.

The classical input: mixed tensor powers

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Thus one studies the spaces

$$\text{Hom}_{GL_n}([r, s]_n, [p, q]_n)$$

and their behavior when n is large relative to r, s, p, q .

By rigidity in $\text{Rep } GL_n$, there is a canonical identification

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The dependence on n is simple: every closed loop produced in a composition contributes the scalar n .

Walled Brauer diagrams: the wall rule

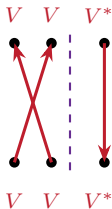
The stable Hom spaces are described by **walled Brauer diagrams**. A diagram for a morphism $[r, s] \rightarrow [p, q]$ has a vertical wall separating the contravariant side (r copies of V at the bottom and p at the top, left of the wall) from the covariant side (s copies of V^* at the bottom and q at the top, right of the wall).

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- A strand which does not cross the wall goes from the bottom row to the top row on the left and from the top row to the bottom row on the right.

no wall crossing



crossing: top-top



crossing: bottom-bottom

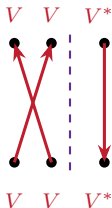


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- A strand which does not cross the wall goes from the bottom row to the top row on the left and from the top row to the bottom row on the right.
- A strand which crosses the wall bends back: it connects two top endpoints (oriented to the left), or two bottom endpoints (oriented to the right).

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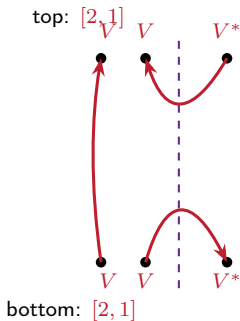
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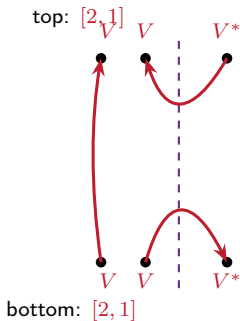
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The diagram encodes a matching together with the orientation rule; there is no difference between over-crossings and under-crossings.

Composition and the loop scalar

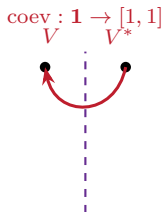
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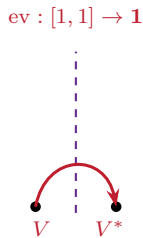
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The basic example is the composition

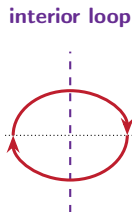
$$\mathbf{1} \xrightarrow{\text{coev}} [1, 1] \xrightarrow{\text{ev}} \mathbf{1}, \quad \text{ev} \circ \text{coev} = n \text{ id}_{\mathbf{1}}.$$



$\circ$



$=$



remove loop; multiply by n

Polynomiality and interpolation

For fixed p, q, r, s and large n , the composition gives bilinear maps

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This gives the diagrammatic interpolation of the tensor category generated by the defining representation of GL_n and its dual.

The category $\widetilde{\text{Rep}}GL_t$

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The tensor product is $[p, q] \otimes [r, s] = [p + r, q + s]$ (**putting diagrams next to each other and moving the V to left and V^* of the right**) and duality is $[r, s]^* = [s, r]$ (**rotating diagrams 180°**). The unit object is $\mathbf{1} = [0, 0]$. The braiding is swapping two diagrams which are next to each other.

Symmetric pseudotensor categories

Definition. A **symmetric pseudotensor category** over $\mathbb{k}$ is a Karoubian (i.e., additive idempotent complete) $\mathbb{k}$ -linear rigid symmetric monoidal category $\mathcal{C}$ such that

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The category $\operatorname{Rep} GL_t := \operatorname{Kar}(\widetilde{\operatorname{Rep} GL_t})$ (the additive idempotent completion of $\widetilde{\operatorname{Rep} GL_t}$) is naturally a symmetric pseudotensor category.

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Lemma. If $W_{r,s}(t)$ is semisimple for all r, s , then $\text{Rep} GL_t$ is a semisimple tensor category.

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Namely, $V_{\lambda,\mu}$ interpolates the irreducible GL_n -module with highest weight

$$(\lambda_1, \dots, \lambda_r, 0, \dots, 0, -\mu_s, \dots, -\mu_1).$$

for $n \geq r + s$.

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Thus

$$\ell(V^{\otimes d}) \geq \sqrt{d!},$$

which is faster than exponential growth.

Tensor ideals

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The **quotient** $\mathcal{C}/\mathcal{I}$ is the symmetric pseudotensor category which has the same objects as $\mathcal{C}$ and morphism spaces

$$\text{Hom}_{\mathcal{C}/\mathcal{I}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y) / \mathcal{I}(X, Y).$$

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The negligible morphisms form a tensor ideal, denoted $\mathcal{N} \subseteq \mathcal{C}$.

Semisimplicity of the negligible quotient

Lemma. (Benson) Suppose that for every object $X \in \mathcal{C}$, every nilpotent endomorphism $a \in \text{End}_{\mathcal{C}}(X)$ satisfies $\text{Tr}(a) = 0$ (**trace zero property**).

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(ii) is straightforward.

Corollary If $\mathcal{C}$ has trace zero property then the quotient $\mathcal{C}/\mathcal{N}$ is a semisimple symmetric tensor category, with simples being the indecomposables of $\mathcal{C}$ of nonzero dimension.

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Consequently,

$$(\operatorname{Rep} GL_t)^{ss} \simeq \operatorname{Rep}_{\mathbb{C}} GL_n.$$

Abelian envelopes and universal property

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This category is not semisimple. It contains $\text{Rep } GL_t$ and retains the universal property of being generated by an object V admitting a dual and $\dim(V) = t$: Any **faithful** symmetric monoidal functor from $\text{Rep } GL_t$ to a symmetric tensor category uniquely factors through $\text{Rep}^{ab} GL_t$.

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There is a boring class of ultrafilters: given $x \in I$, we have $A \in \mathcal{F}_x$ iff $x \in A$. Such ultrafilters are called **principal**; they correspond to maximal ideals $\mathfrak{m}_x$, $x \in I$.

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- if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$;
- if $A \in \mathcal{F}$ and $A \subseteq B \subseteq I$, then $B \in \mathcal{F}$;
- for every $A \subseteq I$, exactly one of A and $I \setminus A$ lies in $\mathcal{F}$; in particular, $I \in \mathcal{F}$.

Alternatively, one may say that an ultrafilter on I is a homomorphism $\chi : \text{Fun}(I, \mathbb{F}_2) \rightarrow \mathbb{F}_2$, i.e., a maximal ideal ($\text{Ker}\chi$) in this ring. Then $A \in \mathcal{F}$ iff $\chi(1_A) = 1$.

There is a boring class of ultrafilters: given $x \in I$, we have $A \in \mathcal{F}_x$ iff $x \in A$. Such ultrafilters are called **principal**; they correspond to maximal ideals $\mathfrak{m}_x$, $x \in I$. Other ultrafilters are called **non-principal**. It is clear that non-principal ultrafilters can exist only on infinite sets.

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Ultraproducts of sets and algebraic structures

Let $\mathcal{F}$ be a non-principal ultrafilter on a set I , and let $\{X_i\}_{i \in I}$ be a collection of sets.

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The **ultraproduct**

$$\prod_{i \in I}^{\mathcal{F}} X_i$$

is the set of functions

$$x : A_x \rightarrow \bigsqcup_{i \in I} X_i, \quad x(i) = x_i \in X_i,$$

defined on some domain $A_x \in \mathcal{F}$, modulo the equivalence relation

$$x \sim y \iff \{i \in A_x \cap A_y \mid x_i = y_i\} \in \mathcal{F}.$$

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Thus, unlike the usual product, the ultraproduct can be non-empty even if some of the factors are empty.

First-order structures and first-order properties are compatible with this construction. For example, ultraproducts of groups, rings, modules, and fields are again groups, rings, modules, and fields.

Ultraproducts of sets and algebraic structures

More precisely, a property P is said to hold **for $\mathcal{F}$ -almost all i** if

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By Łoś's theorem, any first-order property which holds for $\mathcal{F}$ -almost all i holds for the ultraproduct

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For example, consider

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Then K is an algebraically closed field of characteristic zero, since these are first-order properties and all factors have them.

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Also K has cardinality continuum. Hence, by Steinitz's theorem,

$$K \cong \mathbb{C}.$$

Ultraproducts of sets and algebraic structures

As another example, let I be the set of primes and consider

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This is an algebraically closed field of cardinality continuum.

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What is its characteristic? For every fixed prime ℓ , the characteristic is not ℓ , since $\text{char } \overline{\mathbb{F}}_p \neq \ell$ for $\mathcal{F}$ -almost all primes p (in fact, all but one). Hence

$$\text{char } L = 0.$$

Therefore, again by Steinitz's theorem,

$$L \cong \mathbb{C}.$$